10.30 The Photon: From Coulomb Gauge to Covariant Gauges

Srednicki §§55–58 and 62 do most of LecNote21–22 in the course's notation. What he leaves out answers these questions:

- "What does 'shifting by const allowed' (LecNote21 p. 5) leave free?"
- "Which $\varepsilon_\pm$ is helicity $+1$? What is the '. . . ' on LecNote21 p. 9?"
- "When may $g^{\mu\nu}$ replace $\sum \varepsilon\varepsilon^*$ (LecNote23 p. 3)?"
- "Where does $-\frac{1}{2\xi}(\partial\cdot A)^2$ (LecNote22 p. 5) come from?"
- "Why is $e\bar\Psi\gamma^\mu\Psi A_\mu$ not gauge invariant (LecNote22 p. 11) if $J\cdot A$ is?"

Conventions are Srednicki's: $g=\mathrm{diag}(-1,+1,+1,+1)$, $A^\mu=(\varphi,\mathbf{A})$, $A_\mu\to A_\mu-\partial_\mu\Gamma$ (54.18), $A(x)=\int\frac{d^4k}{(2\pi)^4}e^{ikx}\tilde A(k)$, ε^*_λ beside a_λ in (55.11), $D_\mu=\partial_\mu-ieA_\mu$ (58.7) with $e=-0.30$ (58.1). This is LecNote22 p. 12's $\partial_\mu+ieQA_\mu$ at $Q_e=-1$: charge $q=-eQ$ means $D_\mu=\partial_\mu-iqA_\mu$, vertex $iq\gamma^\mu$ (note 22-E5).

Coulomb gauge: residual freedom, helicity, LSZ

Srednicki's projector (55.4) is the gauge transformation with $\Gamma=\nabla^{-2}\nabla\cdot\mathbf{A}=-\int d^3y\,\nabla\cdot\mathbf{A}(\mathbf{y},t)/4\pi|\mathbf{x}-\mathbf{y}|$:

$$A_i-\nabla_i\Gamma=\Big(\delta_{ij}-\frac{\nabla_i\nabla_j}{\nabla^2}\Big)A_j. \tag{10.30.1}$$

A further Γ' keeps $\nabla\cdot\mathbf{A}=0$ if $\nabla^2\Gamma'=0$: $\Gamma'=\mathbf{c}\cdot\mathbf{x}$ shifts $\mathbf{A}$ by the constant $-\mathbf{c}$, LecNote21 p. 5's "shifting by const". If $\mathbf{A}$ must vanish at infinity, so must the harmonic $\nabla_i\Gamma'$, which then vanish everywhere (maximum principle). Then $\Gamma'=c(t)$ shifts φ by $\dot c$ (54.7), and $\varphi\to0$ at infinity (55.7) makes c a constant: a global phase (58.15). A Proca mass leaves no freedom: the field equation gives $m^2\partial\cdot A=\partial\cdot J$, and the residue $g^{\mu\nu}+k^\mu k^\nu/m^2$ of (10.20.5) projects onto three polarizations.

With §10.3's sign, a rotation by θ about $\hat{\mathbf{z}}$ ($\omega_{12}=-\theta$) is $e^{-i\theta J_z}$ both on states ($J_z=M^{12}$) and on vector components ($J_z=S^{12}_V$, $(J_z)^1{}_2=-i=-(J_z)^2{}_1$). It turns $(0,1,0,0)$ into $(0,\cos\theta,\sin\theta,0)$, counterclockwise as in quantum mechanics, and

$$J_z\,\varepsilon^*_\pm=\pm\varepsilon^*_\pm,\qquad \varepsilon^*_\pm=\tfrac{1}{\sqrt2}(0,1,\pm i,0)\quad\text{for }\mathbf{k}\parallel\hat{\mathbf{z}}\text{ (55.12)}. \tag{10.30.2}$$

The star belongs: $\langle0|A^\mu(0)|k,\lambda\rangle=\varepsilon^{\mu*}_\lambda$ (55.11) and $U^{-1}A^\mu U=R^\mu{}_\nu A^\nu$ (§10.4), so the helicity of $a^\dagger_\lambda$ is the J_z eigenvalue of ε^*_λ: $+1$ for $a^\dagger_+$ (Srednicki, after (55.23)).

The '. . . ' on LecNote21 p. 9 (Srednicki's "line by line" before (56.3)) is ∂_0 of (56.2), with $\partial_0 e^{-ikx} = i\omega e^{-ikx}$ and $\omega^2 e^{-ikx} = -\nabla^2 e^{-ikx}$ moved onto $\mathbf{A}$ by parts:

$$\partial_0 a_\lambda(\mathbf{k},t) = i\varepsilon_\lambda \cdot \int d^3x\, e^{-ikx}(\partial_0^2 + \omega^2)\mathbf{A} = i\varepsilon_\lambda \cdot \int d^3x\, e^{-ikx}(-\partial^2)\mathbf{A}. \qquad (10.30.3)$$

Its $\int dt$ is $a_\lambda(+\infty) - a_\lambda(-\infty)$; (56.1) gives $a^\dagger_\lambda(-\infty) - a^\dagger_\lambda(+\infty)$ the same way, with $i\varepsilon^*_\lambda$ and e^{+ikx}. With $\varepsilon^0 \equiv 0$ these are (56.4) and (56.3): an incoming photon brings $\varepsilon^{\mu *}$, an outgoing one ε^μ (§58 rule 9), a convention fixed by (55.11).

Four polarizations, and when $\sum \varepsilon\varepsilon^* \to g$

Srednicki (56.19)–(56.28) takes the Coulomb-gauge propagator (56.16) to Feynman gauge. His (56.24), read backwards with $n = (1,\mathbf{0})$, $l = (0, \mathbf{k}/|\mathbf{k}|)$ (his $\hat{t}$, $\hat{z}$), gives $g^{\mu\nu}$ four polarizations,

$$g^{\mu\nu} = \sum_{\lambda=\pm} \varepsilon^{\mu *}_\lambda \varepsilon^\nu_\lambda - n^\mu n^\nu + l^\mu l^\nu, \qquad (10.30.4)$$

the time-like one with the wrong sign: LecNote22 p. 1's alarm. Between conserved currents $l \cdot \tilde{J} = k^0 \tilde{J}^0/|\mathbf{k}|$, so the extra pair gives $(-|\tilde{J}^0|^2 + |l\cdot\tilde{J}|^2)/(k^2 - i\epsilon) = -|\tilde{J}^0|^2/\mathbf{k}^2$, the instantaneous Coulomb term of (56.16), with no pole at $k^2 = 0$. Only the two transverse photons propagate.

On shell, $-nn + ll$ is a sum of terms with k^μ or k^ν, dropped in (59.10)–(59.17) because $k_\mu \mathcal{M}^\mu = 0$. The Ward identity (§67) gives that for the full amplitude only: in $e^+e^- \to \gamma\gamma$, (59.1), k_1 in place of ε_1 cancels between the t- and u-channel diagrams, not in either alone (Problem 67.2). So g may replace the sums only if all four terms of $|T_t + T_u|^2$ are kept. At $m = 1$, CM beam energy 1.7 and 0.83 rad between $\mathbf{k}_1$ and $\mathbf{p}_1$ ($s = 11.56$, $t = -1.625$), the t-channel term alone, $\frac{1}{4}\sum|T_t|^2$, is $7.145\,e^4$ summed over the transverse polarizations of the CM frame (another frame gives another value) and $7.169\,e^4$ with g; the full $\langle|T|^2\rangle$ is $9.393\,e^4$ both ways (§10.29).

The path integral: gauge orbits and Faddeev–Popov for $U(1)$

The k^μ direction Srednicki drops from $\mathcal{D}A$ (his p. 344, after (57.8)) is the gauge orbit $A \to A - \partial\Gamma$. Along it F is fixed and $\int d^4x\, J\cdot A$ changes by $\int d^4x\, \Gamma\, \partial\cdot J$, so the orbit integral $\int \mathcal{D}\Gamma\, e^{i\int \Gamma\, \partial\cdot J} \propto \prod_k \delta(k_\mu \tilde{J}^\mu(k))$ is an infinite constant for conserved J, cancelling in $Z_0(J)/Z_0(0)$, and zero otherwise: a photon cannot couple to a non-conserved source.

Srednicki fixes the gauge with a δ-function only for nonabelian fields, (71.8)–(71.23). For $U(1)$, $\partial\cdot A^\Gamma = \partial\cdot A - \partial^2\Gamma$ ($A^\Gamma \equiv A - \partial\Gamma$), so for any fixed

$\omega(x)$

$$1 = \det(-\partial^2) \int \mathcal{D}\Gamma \; \delta\big[\partial^\mu A^\Gamma_\mu - \omega\big], \tag{10.30.5}$$

with an A-independent Jacobian: (71.18) at $D_\mu \to \partial_\mu$, so no ghost couples (Srednicki p. 423). Insert it in $Z = \int \mathcal{D}A\, e^{iS}$ and shift $A \to A + \partial\Gamma$ (S and $\mathcal{D}A$ are unchanged for conserved J): the orbit volume $\int \mathcal{D}\Gamma$ factors out, and $Z \propto \int \mathcal{D}A\, \delta[\partial \cdot A - \omega]\, e^{iS}$ for every ω. Averaging over ω with weight $\exp[-\frac{i}{2\xi}\int d^4x\, \omega^2]$ (71.21) sets $\omega = \partial \cdot A$ and adds $-\frac{1}{2\xi}(\partial_\mu A^\mu)^2$ to $\mathcal{L}$: the kernel $k^2 g^{\mu\nu} - k^\mu k^\nu$ of (57.3) becomes $k^2 g^{\mu\nu} - (1 - \frac{1}{\xi}) k^\mu k^\nu$. Its inverse is the R_ξ propagator $[g_{\mu\nu} - (1-\xi)k_\mu k_\nu / k^2]/(k^2 - i\epsilon)$, (62.5), (62.9) (Problem 62.1), whose ξ term drops out on conserved currents (LecNote23 p. 6).

Gauge invariance with a dynamical current

The field equations of (58.6) are $E = 0$ and $\bar{E} = 0$, with $E \equiv (-i\not{\partial} - e\not{A} + m)\Psi$ and $\bar{E} \equiv i\partial_\mu \bar{\Psi}\gamma^\mu - e\bar{\Psi}\not{A} + m\bar{\Psi}$. For every Ψ,

$$\partial_\mu(\bar{\Psi}\gamma^\mu\Psi) = i\big(\bar{\Psi}E - \bar{E}\Psi\big), \tag{10.30.6}$$

which Srednicki states in words before (58.3). So $J^\mu = e\bar{\Psi}\gamma^\mu\Psi$ is conserved only on solutions, unlike the external J of LecNote21 p. 3, and with Ψ fixed, $A \to A - \partial\Gamma$ changes S by $\int d^4x\, \Gamma\, \partial \cdot J \neq 0$ (LecNote22 p. 11). Letting $\Psi \to e^{-ie\Gamma}\Psi$ as well changes $i\bar{\Psi}\not{\partial}\Psi$ by $e\bar{\Psi}\gamma^\mu\Psi\, \partial_\mu\Gamma = -\Gamma\, \partial \cdot J +$ (total derivative), which cancels it for every Ψ, as the path integral needs. $\mathcal{D}\bar{\Psi}\mathcal{D}\Psi$ is invariant too (opposite phases), so the steps above go through with Ψ transformed along with A: the $U(1)$ case of the careful argument that Srednicki p. 347 defers to §71. The rest is (58.3)–(58.16).